Akash Gogate

Princeton, NJ ● 609-751-2597 ● akash.gogate@gmail.com ● agogate@wisc.edu ● linkedin.com/in/akashgogate ● github.com/AkashGogate

EDUCATION

University of Wisconsin at Madison *Graduating: May 2027*
Bachelor of Science in Computer Science and Biology ● Dean's List Fall 2024, Spring 2025, Fall 2025 ● 3.9 GPA
Relevant Coursework: Deep Learning, Artificial Intelligence, Advanced Algorithms, Linear Algebra, Differential Equations, Data Structures, Cryptography
South Brunswick High School
Relevant Coursework: Computer Science Capstone, AP Statistics, AP Biology, Android Application Development, Virtual Reality, Game Development, Web Development

TECHNOLOGIES

Languages: Python, R, TypeScript, Node.js, Java, JavaScript, C++, SQL, Linux/Bash
Bioinformatics & Genomics: Scanpy, Seurat, CARD, Bioconductor, DESeq2, scVI, scANVI, scGen, AmortizedLDA, scRNA-seq, NGS, Next-Generation Sequencing, Spatial Transcriptomics, Computational Biology, Gene Expression Analysis, Differential Expression, Pathway Analysis, Drug Discovery, Precision Medicine, Single-Cell Analysis, Pipeline Development, EHR Integration (HL7/FHIR), Clinical Validation
ML & AI: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Sentence-Transformers, NLP, RAG, Semantic Similarity Search, Random Forest, KNN, Statistical Modeling, Cross-Validation
Systems & Infra: Kafka, Kubernetes, Docker, MongoDB, SQLite, FastAPI, REST APIs, GitHub Actions, CI/CD, Git, pytest, Jupyter Notebook

EXPERIENCE

The Kendziorski Lab at the University of Wisconsin-Madison *September 2025 - Present*
Student Research Intern
- Built agentic pipeline optimization framework for spatial transcriptomics and scRNA-seq (NGS) analysis, autonomously tuning models across scVI, scANVI, scGen, and AmortizedLDA to **cut GPU costs by 300%**; deployed with clinical validation in a production genomics environment serving glioblastoma gene therapy workflows at UW-Madison Medical Center
- Engineered TransferAgent module automating citation mapping, peer-reviewed literature synthesis, and cross-paper validity scoring for clinical research review, **cutting review time by ~60%** and enabling faster iteration on drug discovery and precision medicine findings

Inspirit AI Research Internship *Sep 2022 - Mar 2023*
Machine Learning Research Intern
- Built skin cancer detection pipeline using OpenCV preprocessing and feature extraction fed into scikit-learn classifiers (Decision Tree, KNN) achieving **93% classification accuracy** via hyperparameter tuning and cross-validation; published as peer-reviewed paper through Inspirit AI
- Separately trained a Random Forest classifier on miRNA sequences achieving **95% predictive accuracy ($p < 0.05$)** for cancer cell likelihood prediction, validated via stratified k-fold cross-validation across multiple data splits

Leidos *May 2025 - Present*
Software Engineer Intern
- Designed Kafka event-driven data pipeline persisting to MongoDB, achieving **low-latency, fault-tolerant streaming** with full system decoupling; deployed on Kubernetes via **GitHub Actions CI/CD** with comprehensive unit test coverage across all components
- Optimized satellite observation scheduling via memoization + DP in Python, **reducing constraint violations** while maintaining full mission objective alignment under real-time operational constraints

TECHNICAL PROJECTS

Lotus Health (https://github.com/AkashGogate/lotus-health) *March 2026*
- Built a clinical risk scoring engine computing 5-year disease cost paths across **45M insurance records** on a **1,080-node ICD-10 comorbidity graph** via SQL queries over MEPS and CMS datasets; output integrates with HIPAA-compliant, EHR-adjacent clinical decision support workflows as a validated patient-level risk score
- Implemented RAG pipeline with semantic similarity search over ICD-10 graph as a vector database to match patient-described symptoms to disease risk profiles; hybrid Groq Llama + Python NLP architecture keeps all financial and clinical outputs hallucination-free and auditable

miRcore Cancer Detection Research *Jul 2021*
- Trained Random Forest classifier in R on micro-RNA expression sequences achieving **95% predictive accuracy ($p < 0.05$)** for cancer cell likelihood prediction, validated via stratified k-fold cross-validation across multiple data splits

Computer Vision Hand Tracking System (https://github.com/AkashGogate/HandOrientationTracking) *March 2024*
- Built real-time gesture recognition and joint angle tracking pipeline in Python using OpenCV and Mediapipe; outputs biomechanical movement data for clinical motion capture, rehabilitation monitoring, and quantitative sports performance analysis

USTA Tournament Explorer (https://github.com/AkashGogate/myUSTA) *Jun 2022*
- Built Android app consuming USTA's GraphQL API to surface local tournament listings and registration data via RESTful API integration; demonstrates end-to-end mobile app development from API design to user-facing health-data presentation

LEADERSHIP

Princeton Racket Club *May 2024 - Aug 2024*
Tennis Coach and Tournament Director
- Directed tournament logistics; coached athletes across skill levels in cross-functional environment; managed end-to-end client communications

Tennis Racket Stringing Services *Jan 2019 - Present*
- Founded and scaled to **50+ clients**; manage all operations, inventory, and client consultations